

4_13_vacc_rate

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2026-04-13

Loading relevant packages and files

library addition

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2    3.5.2      v tibble    3.3.0
## v lubridate  1.9.4      v tidyr     1.3.1
## v purrr      1.0.4
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag()     masks stats::lag()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(lubridate)
```

```
library(MMWRweek) # for MMWRweek2Date
```

```
library(gtsummary)
```

```
library(flextable)
```

```
##
```

```
## Attaching package: 'flextable'
```

```
##
```

```
## The following object is masked from 'package:gtsummary':
```

```
##
```

```
##     continuous_summary
```

```
##
```

```
## The following object is masked from 'package:purrr':
```

```
##
```

```
##     compose
```

```
library(summarytools)
```

```
##
```

```
## Attaching package: 'summarytools'
```

```
##
```

```
## The following object is masked from 'package:tibble':
```

```
##
```

```
##     view
```

```
library(fitdistrplus)
```

```
## Loading required package: MASS
##
## Attaching package: 'MASS'
##
## The following object is masked from 'package:gtsummary':
##
##     select
##
## The following object is masked from 'package:dplyr':
##
##     select
##
## Loading required package: survival
library(webshot2)
library(coin)
```

paths

```
out_dir <- "~/Documents/Documents/Yellow Fever/YF_scoping_review/fig"
src_path <- "~/Documents/Documents/Yellow Fever/YF_scoping_review/src"
data_path <- "~/Documents/Documents/Yellow Fever/YF_scoping_review/data"

file_path <- file.path(data_path, "YF_RVCs.csv")
```

file reading

```
df <- read_csv(file_path)

## Rows: 112 Columns: 16
## -- Column specification -----
## Delimiter: ","
## chr (13): Country, outbreak.identifier, react.vac.start.date.one, react.vac....
## dbl (2): reac.target.pop.one, reac.vac.perc.one
## num (1): reac.vac.one
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

Cleaning/column creation

date creation

```
# parse dates (M/D/YYYY format)
df <- df %>%
  mutate(
    start_date = mdy(react.vac.start.date.one),
    end_date   = mdy(react.vac.end.date.one)
  ) %>%
  filter(vacc.marker != "DRC_2014_1_pv") #because there is an aggregate estimate that works better
```

establish vaccination types

```
df <- df %>%
  mutate(
    # split vacc.marker
    base_marker = str_remove(vacc.marker, "_ (pv|im|admin)$"),
    type = case_when(
      str_detect(vacc.marker, "_pv$") ~ "pv",
      str_detect(vacc.marker, "_im$") ~ "im",
      str_detect(vacc.marker, "_admin$") ~ "admin",
      TRUE ~ "admin" # treat null as admin
    ),

    # priority: im/pv > admin
    priority = case_when(
      type %in% c("im", "pv") ~ 1,
      TRUE ~ 2
    )
  )
```

establish geography markers

```
geo_cols <- c(
  "Geographic.marker.1",
  "Geographic.marker.2",
  "Geographic.marker.3"
)

df <- df %>%
  unite("geo_id", all_of(geo_cols), sep = "|", na.rm = TRUE, remove = FALSE)
```

creating columns for completing vaccination coverage information

```
df_clean <- df %>%
  mutate(
    reac.vac.perc.one = if_else(
      is.na(reac.vac.perc.one) & !is.na(reac.vac.one) & !is.na(reac.target.pop.one),
      reac.vac.one / reac.target.pop.one * 100,
      reac.vac.perc.one
    ),
    reac.target.pop.one = if_else(
      is.na(reac.target.pop.one) & !is.na(reac.vac.one) & !is.na(reac.vac.perc.one),
      (reac.vac.one / reac.vac.perc.one) * 100,
      reac.target.pop.one
    ),
    reac.vac.one = if_else(
      is.na(reac.vac.one) & !is.na(reac.target.pop.one) & !is.na(reac.vac.perc.one),
      (reac.vac.perc.one / 100) * reac.target.pop.one,
      reac.vac.one
    )
  )
```

creating and following priority rule

```
df_clean <- df_clean %>%
  mutate(
    completeness = rowSums(!is.na(pick(
      reac.vac.one,
      reac.target.pop.one,
      reac.vac.perc.one
    )))
  ) %>%
  group_by(base_marker, geo_id) %>%
  arrange(desc(completeness), priority, .by_group = TRUE) %>%
  slice_head(n = 1) %>%
  ungroup()

df_clean

## # A tibble: 93 x 23
##   Country outbreak.identifier react.vac.start.date.one react.vac.end.date.one
##   <chr>      <chr>                <chr>                <chr>
## 1 Angola   Angola_2016                2/19/2016                4/30/2016
## 2 Angola   Angola_2016                2/29/2016                4/30/2016
## 3 Angola   Angola_2016                2/29/2016                4/30/2016
## 4 Angola   Angola_2016                3/28/2016                4/30/2016
## 5 Angola   Angola_2016                3/24/2016                4/30/2016
## 6 Angola   Angola_2016                3/10/2016                4/30/2016
## 7 Angola   Angola_2016                3/28/2016                4/30/2016
## 8 Angola   Angola_2016                3/14/2016                4/30/2016
## 9 Angola   Angola_2016                3/28/2016                4/30/2016
## 10 Angola  Angola_2016                3/24/2016                4/30/2016
## # i 83 more rows
## # i 19 more variables: react.vac.start.date.two <chr>,
## #   react.vac.end.date.two <chr>, geo_id <chr>, Geographic.marker.1 <chr>,
## #   Geographic.marker.2 <chr>, Geographic.marker.3 <chr>, vacc.marker <chr>,
## #   reac.vac.one <dbl>, reac.target.pop.one <dbl>, reac.vac.perc.one <dbl>,
## #   source.one <chr>, source.two <chr>, source.three <chr>, start_date <date>,
## #   end_date <date>, base_marker <chr>, type <chr>, priority <dbl>, ...
```

mark by country and year

```
df_clean <- df_clean %>%
  mutate(
    country = str_extract(base_marker, "[^_]+"),
    year     = as.numeric(str_extract(base_marker, "\\d{4}"))
  )
```

creating the dataframe of relevent vaccination information

```
campaign_df <- df_clean %>%
  group_by(base_marker) %>%
  summarise(
    outbreak.identifier = first(outbreak.identifier),
    country = first(country),
```

```

year = first(year),
start = min(start_date, na.rm = TRUE),
end   = max(end_date, na.rm = TRUE),
duration = as.numeric(end - start + 1),
  target_pop = sum(reac.target.pop.one),
vaccinated_pop = sum(reac.vac.one),
n_rows = n(),
coverage = sum(reac.vac.one, na.rm = TRUE) /
            sum(reac.target.pop.one, na.rm = TRUE) * 100,
rate = sum(coverage) / sum(duration),
.groups = "drop"
)

```

Classify pre vs post-Angola Campaigns

```

cutoff <- as.Date("2017-02-14")

campaign_df <- campaign_df %>%
  mutate(
    period = if_else(start < cutoff, "pre_angola", "post_angola")
  )

```

Results

the pooled weighted estimate and standard error

```

ratio_var <- campaign_df %>%
  filter(!is.na(coverage), !is.na(duration)) %>%
  summarise(
    n = n(),
    Xbar = mean(duration),
    Ybar = mean(coverage),

    r_hat = sum(coverage) / sum(duration),

    s2_X = var(duration),
    s2_Y = var(coverage),
    s_XY = cov(duration, coverage, use = "complete.obs"),

    var_ratio = (1 / (n * Xbar^2)) *
      (r_hat^2 * s2_X + s2_Y - 2 * r_hat * s_XY),

    se_ratio = sqrt(var_ratio)
  )

ratio_var

## # A tibble: 1 x 9
##       n  Xbar  Ybar r_hat  s2_X  s2_Y  s_XY var_ratio se_ratio
##   <int> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>   <dbl>   <dbl>
## 1    29  11.6  92.1  7.93  235.  183. -18.3     3.89     1.97

```

the pooled weighted estimate and standard error by period

```
tot = nrow(campaign_df %>%
  filter(!is.na(coverage), !is.na(duration)))

var_ratio_by_period <- campaign_df %>%
  filter(!is.na(coverage), !is.na(duration)) %>%
  group_by(period) %>%
  summarise(
    n = n(),
    perc = round(n()/tot *100,2),
    Xbar = mean(duration),

    r_hat = sum(coverage) / sum(duration),

    s2_X = var(duration),
    s2_Y = var(coverage),
    s_XY = cov(duration, coverage),

    var_ratio = (1 / (n * Xbar^2)) *
      (r_hat^2 * s2_X + s2_Y - 2 * r_hat * s_XY),

    se_ratio = sqrt(var_ratio),
    .groups = "drop"
  )
var_ratio_by_period
```

```
## # A tibble: 2 x 10
##   period      n perc  Xbar r_hat  s2_X  s2_Y  s_XY var_ratio se_ratio
##   <chr>    <int> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>    <dbl>    <dbl>
## 1 post_angola    18 62.1  9.22  9.76  11.4 252.  -9.75     0.996     0.998
## 2 pre_angola    11 37.9 15.5   6.14 611.  64.8 -58.5     8.97     3.00
```

T test

```
t_score_a = (var_ratio_by_period[1,5] - var_ratio_by_period[2,5]) / sqrt(var_ratio_by_period[1,10] + var_ratio_by_period[2,10])
n = tot - 2
2 * pt(-as.numeric(t_score_a), df = n)

## [1] 0.08137093
```

histogram of pooled reactive vaccination

```
pub_table <- campaign_df %>%
  arrange((start)) %>%
  filter(!is.na(rate)) %>%
  transmute(
    `Start date` = start,
    `End date` = end, # force integer, character (no commas)
    Country = country,
    `Target Population` = round(target_pop,0),
    `Coverage (%)` = round(coverage,2),
    `Vaccination rate` = round(rate,2)
```

```

)

pub_table = pub_table %>%
  relocate(Country, .before = `Start date`)

ft_pub <- flextable(pub_table) %>%
  theme_booktabs() %>%
  bold(part = "header") %>%
  align(j = 1, align = "left", part = "body") %>%
  align(j = 2:ncol(pub_table), align = "center", part = "all") %>%
  autofit() %>%
  fontsize(size = 10, part = "all") %>%
  set_header_labels(
    Country = "Country",
    `Start date` = "Start date",
    `End date` = "End date",
    `Target Population` = "Target Population",
    `Coverage (%)` = "Coverage (%)",
    `Vaccination rate` = "Vaccination rate (% per day)"
  )

png_file <- file.path(out_dir, "table_2S.png")
docx_file <- file.path(out_dir, "table_2S.docx")

save_as_image(ft_pub, path = png_file, expand = 8, zoom = 2)

## [1] "~/Documents/Documents/Yellow Fever/YF_scoping_review/fig/table_2S.png"

save_as_docx(ft_pub, path = docx_file)

ft_pub

## Warning: fonts used in `flextable` are ignored because the `pdflatex` engine is
## used and not `xelatex` or `lualatex`. You can avoid this warning by using the
## `set_flextable_defaults(fonts_ignore=TRUE)` command or use a compatible engine
## by defining `latex_engine: xelatex` in the YAML header of the R Markdown
## document.

```

Country	Start date	End date	Target Population	Coverage (%)	Vaccination rate per day)
Uganda	2011-01-20	2011-01-25	905,618	84.90	14.15
Ghana	2011-11-22	2011-11-28	235,082	82.30	11.76
Cameroon	2012-01-23	2012-01-29	1,170,258	93.90	13.41
Chad	2013-02-22	2013-03-02	1,017,295	97.42	10.82
DRC	2013-06-20	2013-06-24	503,426	102.00	20.40
Cameroon	2013-08-27	2013-09-01	663,900	94.00	15.67
Cameroon	2014-01-03	2014-01-09	245,716	101.84	14.55
DRC	2014-05-20	2014-05-24	504,394	105.00	21.00
Angola	2016-02-02	2016-04-30	6,583,216	88.40	0.99

Country	Start date	End date	Target Population	Coverage (%)	Vaccination rate per day)
Uganda	2016-05-19	2016-06-07	668,236	93.93	4.70
DRC	2016-05-26	2016-06-04	3,955,467	106.76	10.68
Nigeria	2017-10-11	2017-10-20	2,023,556	89.09	8.91
Nigeria	2017-12-01	2017-12-10	1,563,005	80.00	8.00
Nigeria	2018-03-10	2018-03-19	1,933,644	88.77	8.88
Nigeria	2018-03-19	2018-03-28	1,343,413	91.00	9.10
Nigeria	2018-08-09	2018-08-18	208,285	74.00	7.40
Ethiopia	2018-10-13	2018-10-19	31,365	99.20	14.17
Ethiopia	2018-11-16	2018-11-23	1,307,908	102.68	12.83
Nigeria	2018-12-18	2018-12-24	1,444,149	120.10	17.16
SouthSudan	2019-03-25	2019-03-29	19,578	102.00	20.40
Nigeria	2019-09-07	2019-09-19	1,529,111	73.12	5.62
SouthSudan	2020-10-23	2020-10-27	704,005	97.50	19.50
Ghana	2021-12-17	2021-12-21	568,467	94.21	18.84
Chad	2022-01-19	2022-01-30	1,117,848	107.00	8.92
Ghana	2022-02-26	2022-03-02	603,394	89.31	17.86
Kenya	2022-07-23	2022-08-01	1,422,600	51.62	5.16
CAR	2022-11-02	2022-11-19	345,920	101.70	5.65
SouthSudan	2024-02-13	2024-02-21	406,699	70.83	7.87
SouthSudan	2024-03-19	2024-03-30	201,570	88.17	7.35